

Bulk heterojunction solar cell.

Bulk heterojunction has an absorption layer consisting of a nano-scale blend of donor and acceptor materials. Bulk heterojunction is most commonly created by forming a solution containing the two components, casting (that is, drop casting and spin coating) and then allowing the two phases to separate, usually with the assistance of an annealing step.

The two components self-assemble into an interpenetrating network connecting the two electrodes. These are

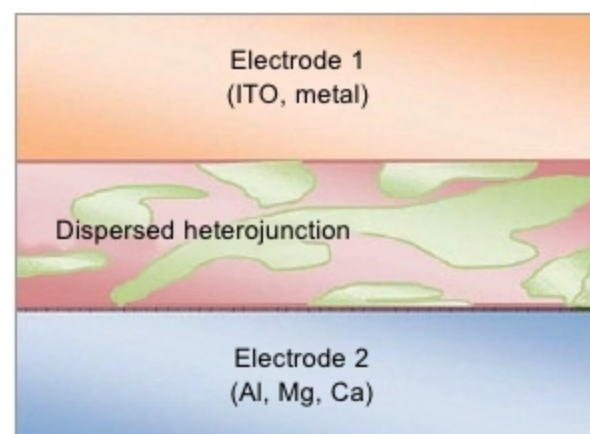


Fig. 6: Structure of a heterojunction solar cell

normally composed of a conjugated molecule-based donor and fullerene-based acceptor. The non-structural morphology of bulk heterojunction tends to be difficult to control but is critical to photovoltaic performance.

After the capture of a photon, electrons move to acceptor domains and are then carried through the device and collected by one electrode. Holes move in the opposite direction and are collected on the other side. If dispersion of the two materials is too fine, it will result in poor charge transfer through the layer.

Most bulk heterojunction cells use two components, although three-component cells have been explored. The third component, a secondary p-type donor polymer, acts to absorb light in a different region of the solar spectrum. This in theory increases the amount of absorbed light. These ternary cells operate through one of three distinct mechanisms: charge transfer, energy transfer or parallel linkage.

In charge transfer, both donors contribute directly to the generation of free charge carriers. Holes pass through only one donor domain before collection at the anode.

In energy transfer, only one donor contributes to the production of holes. The second donor acts solely to absorb light, transferring extra energy to the first donor material.

In parallel linkage, both donors produce excitons independently, which then migrate to their respective donor/acceptor interfaces and dissociate.

What is inside organic solar cells is called bulk heterojunction solar cells. While making organic solar cells, polymer is taken in its semi-solid form and carbon is then mixed in it, so it becomes a semi-solid mixture such as ink of a fountain pen. This ink is made to settle for a considerable amount of time. As the ink settles, it looks exactly like what is shown in Fig. 7. This is how bulk heterojunction is made practically.

Power conversion efficiency (η)

One of the major issues surrounding polymer solar cells is the low power conversion efficiency (PCE) of fabricated cells. To be considered commercially-viable, photovoltaic solar cells must be able to achieve at least ten to fifteen per cent efficiency, which is already much lower than of inorganic photovoltaics. However, due to the low cost of polymer solar cells, ten to fifteen per cent efficiency is commercially-viable.

PCE (η) is proportional to the product of short-circuit current (I_{sc}), open circuit voltage (V_{oc}) and fill factor (FF).

$$\eta = V_{oc} \times I_{sc} \times \text{Fill factor}$$

Recent advances in polymer solar cell performance have resulted from compressing the band gap to enhance short-circuit current while lowering HOMO to increase open-circuit voltage. However, photovoltaic solar cells still suffer from low fill factors (typically below 70 per cent). But, as of 2013, researchers have been able to fabricate photovoltaic solar cells with fill factors of over 75 per cent. Scientists have been able to accomplish this via an inverted

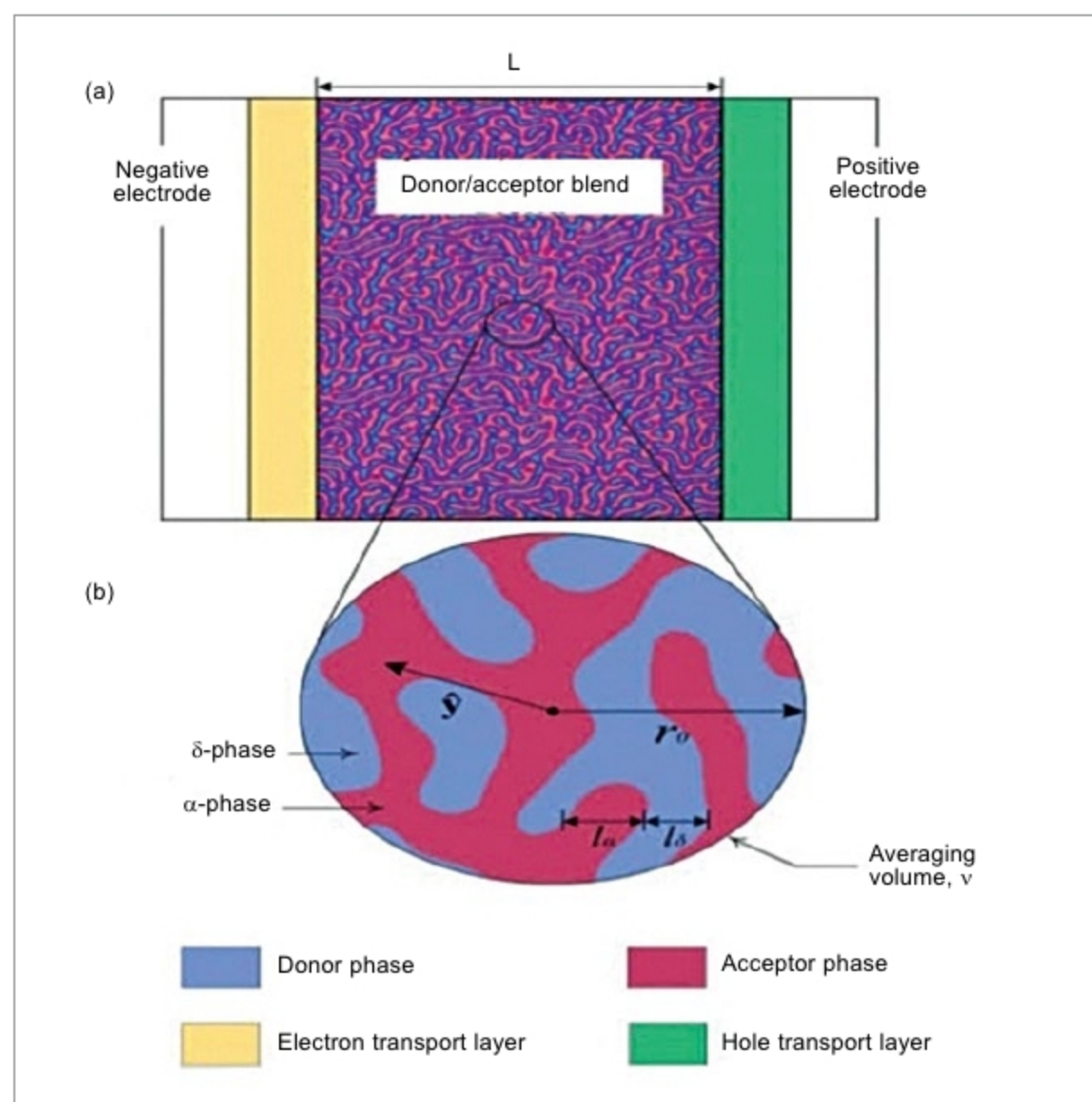


Fig. 7: Internal structure of organic photovoltaic